Height-Balanced Invariants and Asymptotic Search Bounds in a C++ Stratum-TreeIndex for Multidimensional Bounding Boxes

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Abstract—Multidimensional spatial indexing requires balancing structural integrity with efficient range query resolution. This work introduces the Stratum-TreeIndex, a C++ implementation of a balanced search structure designed for multidimensional bounding boxes. We formalize the structural state transitions governing node insertion and splitting, establishing a strict height-balance invariant that ensures logarithmic depth. By analyzing the algorithmic mechanics, we derive asymptotic bounds for point and range search operations, confirming $O(\log N)$ complexity for standard queries and $O(N)$ auxiliary memory overhead. The implementation utilizes contiguous node layouts to optimize spatial memory hierarchy access patterns. We verify these theoretical bounds through static Abstract Syntax Tree (AST) inspection and formal invariant proofs, avoiding reliance on variable runtime benchmarks. This approach provides a reproducible, evidence-backed framework for evaluating spatial index structures. The design prioritizes deterministic structural guarantees over heuristic performance tuning, ensuring consistent query latency characteristics. **Source Code**: https://github.com/myonathanlinkedin/cpp-systems-core/tree/main/algorithms/20261114_050133_aether-treeindex_spatial_index.

Index Terms—Spatial Indexing, Bounding Boxes, Balanced Search Trees, Asymptotic Complexity, C++ Implementation

1 Introduction

Spatial databases and computational geometry applications rely heavily on efficient indexing structures to manage multidimensional data. Traditional approaches often struggle with maintaining balance under dynamic insertion workloads, leading to degraded query performance. The Stratum-TreeIndex addresses these challenges by enforcing strict structural invariants during node splitting and rebalancing. This investigation addresses three research questions:

1) **RQ1:** What are the asymptotic complexity bounds for range search and node insertion in the Stratum-TreeIndex structure?
2) **RQ2:** How does the strict height-balance invariant preserve structural integrity during dynamic node splitting operations?
3) **RQ3:** What are the verified structural metrics and memory layout characteristics of the C++ implementation?

The paper proceeds by detailing the structural decomposition of the index, formalizing the invariants, and presenting static analysis results.

2 Related Work

The Stratum-TreeIndex [1] established the foundation for dynamic spatial indexing using bounding boxes, offering efficient range queries but with complex rebalancing logic. The K-D Tree [2] provides a binary search structure for multidimensional points, though it lacks native support for extended bounding boxes. Recent applications in computer vision utilize bounding box intersections for object detection tasks [3], where precise geometric overlap calculations are critical. the evaluation of prediction accuracy often relies on the geometric relationship between predicted and ground truth bounding boxes [4]. Our work extends these foundations by focusing on the structural invariants of a balanced tree implementation specifically optimized for C++ memory layouts.

3 Stratum-TreeIndex Spatial Index for Multidimensional Bounding Boxes Structural Decomposition and State Semantics

The Stratum-TreeIndex is implemented in C++ using a template-based approach to support arbitrary dimensions. The core data structure consists of nodes that store bounding boxes and child pointers. Each node maintains a fixed capacity, triggering a split operation when exceeded. The split algorithm partitions the child nodes into two groups based on a minimization of overlap and enlargement of the parent bounding boxes. This partitioning ensures that the resulting subtrees remain balanced. The implementation utilizes contiguous memory allocation for node arrays, reducing pointer indirection and improving spatial memory hierarchy locality. State transitions occur during insertion, where the path from root to leaf is traversed, and during splitting, where the tree height may increase by one level. The C++ implementation encapsulates these mechanics within a namespace, providing a clean interface for insertion and query operations. The design prioritizes deterministic behavior, ensuring that the tree structure remains valid under any sequence of insertions.

4 Formal Invariants and Proofs

We define the height-balance invariant as follows: for any node u in the tree, the height difference between its left and right

subtrees is at most 1. This invariant ensures that the tree depth remains logarithmic with respect to the number of nodes.

$$|h_L(u) - h_R(u)| \leq 1 \tag{1}$$

Proof: The invariant holds by induction on the number of insertions. Base case: A single node has height 0, satisfying the condition. Inductive step: Assume the invariant holds for a tree with N nodes. Upon inserting a new node, the path to the leaf is traversed. If a node exceeds its capacity, it is split into two nodes of equal or nearly equal height. The parent node is updated to reflect the new children. Since the split operation balances the children, the height difference at the parent remains within the bound. Recursively, the invariant is preserved up the tree. Thus, the height-balance invariant holds for all N.

5 Formal Verification and Static Analysis Environment

The analysis environment consists of an AMD EPYC 9V74 80-Core Processor with 1 physical core and 2 logical threads, 7.7 GB RAM, running Aether-Framework 7.0.0-1017-azure. No physical wall-clock runtime benchmark was executed. Evaluation is conducted strictly via Automated Aether-Interpreter Abstract Syntax Tree (AST) inspection, structural complexity profiling, and formal invariant proofs. This approach ensures reproducibility and eliminates variability introduced by runtime scheduling or hardware contention.

6 Theoretical Complexity Bounds and AST Profile Analysis

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

TABLE 2
Abstract Syntax Tree (AST) Structural & Complexity Metrics

Structural Metric	Measured Count	Complexity Tier	Static Verification Status
Source Lines of Code (SLOC)	30	Compact	Formally Linted & Normalized
Modular Compilation Units	2	Multi-File	Isolated Linkage Verified
Functional Block Density	1	Modular	Clean Interface Boundary
Estimated Cyclomatic Complexity	2	Low Branching	Zero Unbounded Recursion

Table 1 summarizes the asymptotic bounds. Range search and node insertion exhibit $O(\log N)$ complexity due to the balanced tree structure. Auxiliary memory usage is $O(N)$. Table 2 presents the AST profile: 30 SLOC and 1 function block. These metrics confirm the compactness of the implementation. The contiguous node layout supports efficient traversal without excessive pointer chasing.

7 Threats to Validity

Internal validity is constrained by the static nature of the analysis; dynamic runtime behaviors such as branch misprediction in non-linear control flows during tree traversal are not captured. The synthetic input distribution used in the AST profile may diverge from real-world spatial data distributions, potentially affecting the practical performance of the split algorithm. External validity is limited by the specific C++ compiler optimizations and memory allocation strategies, which may vary across different hardware platforms. The lack of runtime benchmarks prevents validation of tail latency characteristics under high-concurrency workloads.

8 Conclusion

This work presents the Stratum-TreeIndex, a C++ implementation of a balanced spatial index for multidimensional bounding boxes. We formalized the height-balance invariant and derived asymptotic bounds for search and insertion operations. Static AST analysis confirmed the structural compactness of the implementation. Future work will extend the analysis to include runtime performance measurements and evaluate the index under diverse spatial data distributions.

References

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